Quantum Superconducting Devices

Advanced Speaking Script and Slide Guide

Superconductivity and Quantum Materials

April 2026

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1 Presentation Goal and Core Claim

The talk is strongest when it defends one central claim from the first minute to the last:

> Superconductivity becomes a technology platform because it provides both low-loss high-current operation and phase-coherent quantum sensing.

That claim lets you organize the presentation in a clean three-step arc:

1. remind the audience of the minimum theoretical ingredients
2. show that those ingredients generate two device families: superconducting magnets and SQUIDs
3. end with three real applications that look very different but are rooted in the same physics

Core point. Water treatment belongs to the high-field branch. MEG and buried-metal detection belong to the ultra-sensitive sensing branch.

Suggested speech.

When people hear about superconductivity, they often think of something elegant but remote from practical life. My goal today is to show the opposite. The same physics can help clean wastewater, measure brain activity, and detect buried metallic objects. The main idea of this talk is simple: superconductivity becomes useful technology in two complementary ways. First, it can carry very large currents with negligible dissipation, which makes high-field magnets practical. Second, because the superconducting condensate has a coherent phase, tiny changes in magnetic flux can be converted into measurable electrical signals through the Josephson effect. Once those two routes are clear, the applications stop looking disconnected. Water treatment uses the first route, while MEG and buried-metal detection use the second.

2 Recommended Structure for a 20-Minute Talk

This section should function as the operational map of the presentation. The goal is not only to list slide titles, but to explain what each slide must accomplish, what the audience should understand by the end of it, and how each step connects to the next one.

A 20-minute talk works best when it has a visible internal rhythm:

1. an opening that creates curiosity and states the main claim
2. a compact theory block that introduces only the concepts needed later
3. a device block that explains why SQUIDs are the natural bridge from theory to applications
4. three applications that clearly illustrate different engineering goals
5. a final synthesis that reconnects everything to one central message

The audience should never feel that the presentation is becoming a list. At every stage, you should make clear why the next slide follows from the previous one.

2.1 Slide-by-slide map

#	Slide title	Purpose	What you should say	Main visual
1	Title + guiding question	Open with a motivating contrast and state the promise of the talk	Superconductivity may look like an exotic low-temperature phenomenon, but the same physics can enable strong magnets and ultra-sensitive sensors used in very different real applications	Clean collage or graphic motif linking magnets, SQUIDs, and applications
2	One physics, two device families	Establish the narrative structure early	The whole talk is organized around two technological routes: superconducting magnets for strong-field tasks and SQUIDs for ultra-sensitive sensing	Flow diagram from superconductivity to magnets and SQUIDs
3	What makes a superconductor special?	Give the minimum theoretical foundation	Zero resistance is important, but the defining physical signature is the Meissner effect; in type II materials vortices make high-field applications possible	Redrawn Meissner/vortex sketch
4	Macroscopic phase and Josephson effect	Introduce the concept that makes interference devices possible	A superconductor is also described by a coherent phase; once two superconductors are weakly coupled, the Josephson effect turns phase into an observable electrical variable	Josephson junction diagram
5	SQUID as a flux-to-voltage transducer	Introduce the key sensor platform and the bridge to applications	A SQUID is a superconducting interferometer that converts tiny changes in magnetic flux into measurable electrical signals	DC-SQUID loop and periodic response curve
6	Water treatment by superconducting magnetic separation	First application: strong-field branch	Here superconductivity is useful because it provides magnetic force through high field and high gradient, not because of quantum sensing	Three-step HGMS process diagram
7	Why a superconducting magnet helps here	Make the engineering logic explicit	The advantage is not the chemistry itself, but the ability to generate strong fields efficiently and continuously compared with a resistive system	Simple comparison: resistive vs superconducting magnet
8	MEG with SQUID arrays	Second application: mature sensing branch	Here the key resource is extreme sensitivity, because brain magnetic fields are extraordinarily weak and must be read non-invasively	Photo of whole-head MEG system + head/sensor sketch

2.2 Detailed speaking logic by block

Slides 1–2: Opening and narrative setup. These two slides define the whole talk. The first slide should not be treated as a formality. It must already contain the conceptual hook: superconductivity is often introduced as a beautiful but abstract phenomenon, whereas here it will be presented as a platform for real devices. The second slide should immediately simplify the structure for the audience by introducing the two technological branches. If you make this split early, the rest of the presentation becomes easier to follow.

Suggested speech.

At the beginning of the talk, do not start with definitions. Start with a contrast. Present superconductivity as something that may appear remote from daily life, then state that the same physics can be used to clean wastewater, measure brain activity, and detect buried metallic targets. Immediately after that, give the audience a map: one branch of the talk concerns strong-field devices, the other concerns ultra-sensitive sensing devices.

Slides 3–5: Theory and device bridge. This block must remain compact. Its role is not to reproduce a full lecture on superconductivity, but to introduce only the minimum physical concepts needed to make the applications intelligible. The first step is to distinguish superconductivity from simple perfect conductivity through the Meissner effect. The second is to introduce macroscopic phase coherence and the Josephson effect. The third is to show how those concepts become a device through the SQUID. If this block becomes too dense, the applications will feel disconnected from the theory instead of emerging naturally from it.

Suggested speech.

In the theory block, the audience should leave with three clear ideas. First, superconductivity is a genuine phase of matter, not only zero resistance. Second, phase coherence matters because it can control current through the Josephson effect. Third, a SQUID converts that quantum interference into a practical sensor. If those three ideas are clear, the applications will make sense almost automatically.

Slides 6–7: First application, strong-field branch. The first application should be chosen to emphasize that superconductivity is not only about detecting tiny signals. In water treatment, the important point is that the superconducting element is the magnet. That lets you illustrate the first device family in a concrete engineering context. It is useful to separate the application into two slides: one for the process itself and one for the reason a superconducting magnet is advantageous. This prevents the argument from becoming too compressed.

Suggested speech.

Use the water-treatment example to make the first branch concrete. Here the message is simple: superconductivity is useful because it can produce strong magnetic fields with low dissipation. The audience should not confuse this application with SQUID sensing. State explicitly that the superconducting component is the magnet, and that the goal is efficient magnetic separation at high field and high gradient.

Slide 8: Second application, mature ultra-sensitive sensing. MEG is the most mature example in the talk, so it should sound like the point where the audience sees that SQUID technology is not speculative. The emphasis should be on the scale of the magnetic signal, the non-invasive nature of the measurement, and the fact that this is already part of real clinical workflows. This slide is important rhetorically because it gives the sensing branch strong credibility.

Suggested speech.

When you arrive at MEG, make the contrast with water treatment explicit. In the previous case the key resource was magnetic force. Here the key resource is sensitivity to an extremely weak field. That single sentence helps the audience understand why both applications belong in the same presentation even though they look completely different.

Slide 9: Third application, field deployment and low-frequency sensing. The buried-metal or landmine example is useful because it shows that SQUIDs can matter even outside the laboratory and outside the hospital. This gives the talk a broader technological reach. The main thing to explain is the measurement chain: excitation field, induced eddy currents in the target, weak secondary field, then SQUID readout. It is also helpful to stress that this is a more difficult environment, where sensitivity must coexist with noise, portability, and operational constraints.

Suggested speech.

This application should be presented as a change of environment. The point is not just that SQUIDs are sensitive, but that their low-frequency performance can still be valuable when the measurement is performed in a noisy and less controlled setting. That makes the example different from MEG while preserving the same sensing logic.

Slides 10–12: Comparison, realism, and closing. The final three slides should not feel like an appendix. They are where the talk acquires intellectual coherence. The comparison slide prevents the applications from looking like unrelated examples. The limitations-and-outlook slide prevents the presentation from sounding overly optimistic or promotional. The conclusion must then return to the central claim in a compressed, memorable form. A good final slide should not introduce anything new; it should simply make the whole structure visible one last time.

Suggested speech.

In the closing block, the audience should clearly see that the same physics leads to different technologies because the engineering objective changes. One case optimizes magnetic force, another optimizes extreme sensitivity, and another optimizes low-frequency detection in a real environment. Then end by restating the main idea: superconductivity can either create the field or detect the field.

2.3 Timing

- slides 1–2, opening and narrative setup: 2 minutes
- slides 3–5, theoretical essentials and SQUID bridge: 7–7.5 minutes
- slides 6–7, water treatment and superconducting magnetic separation: 3 minutes
- slide 8, MEG with SQUID arrays: 3 minutes
- slide 9, landmine and buried-metal detection: 2.5–3 minutes
- slides 10–12, comparison, limits, and conclusion: 2.5 minutes

Timing. A useful rule is that the theory block should be compact but not rushed. If you spend too long on basic superconductivity, the applications will feel squeezed; if you move too quickly, the audience will not understand why the devices work.

2.4 Practical pacing advice

- slides 1–2 should feel confident and fast, because they only establish the roadmap

- slide 3 is the only place where you should slow down slightly to define the physical state clearly
- slides 4–5 should be explained with one equation each at most, and always with an intuitive sentence after the equation
- slides 6–9 should be concrete, visual, and application-driven rather than theoretical
- slides 10–12 should accelerate slightly again, because by then the audience already has the full picture

Timing. A very common mistake is to use too much time before the first application appears. Ideally, the audience should reach the first concrete application by about minute 9.

2.5 What the audience should remember at the end of this section

By the end of the full 20-minute presentation, the audience should be able to reconstruct the argument in one short chain:

1. superconductivity is more than zero resistance because it includes magnetic expulsion and phase coherence
2. phase coherence and the Josephson effect make interference devices such as SQUIDs possible
3. superconducting technology therefore develops along two main routes: strong-field magnets and ultra-sensitive sensors
4. water treatment, MEG, and buried-metal detection are three concrete examples of those routes
5. the unity of the talk comes from the physics, while the diversity comes from the engineering target

Core point. A good test of the structure is this: if someone remembers only the distinction between strong-field action and ultra-sensitive readout, then the talk has already succeeded.

3 How to Build the Slides

3.1 Practical rules

- one slide, one dominant idea
- keep most slides to two or three bullets
- theory slides should favor clean diagrams over screenshots from lecture notes
- application slides usually work best with one real image plus one simple physical schematic
- use only the equations you will actually explain aloud
- highlight a few memorable numbers instead of many weak details
- make slide titles carry the message, not just the topic label

Slide plan. Treat the slides as visual anchors for your voice, not as a written transcript.

3.2 Visual strategy

- theory: redraw Meissner expulsion, flux vortices, a Josephson junction, and a DC-SQUID loop
- water treatment: use a process flow, not a generic water photo

- MEG: use a real system image, because it immediately signals maturity and clinical relevance
- buried-metal detection: use a technical diagram rather than dramatic field photography
- comparison slide: use a table or icons, not three unrelated images

Visual. If a visual cannot be understood in about five seconds, it is too dense for this talk.

3.3 What to avoid

- long paragraphs on slides
- too many equations in the theory block
- generic titles such as "Application 2"
- emotionally heavy imagery for the landmine section
- a final slide crowded with every conclusion at once

Suggested speech.

The most common mistake in a talk like this is trying to move too much of the script onto the slides. That usually weakens both the visuals and the speaking. A better rule is that each slide should be readable almost instantly, leave some silence around the main figure, and support one claim that you can say in a single sentence.

4 Theory Block

4.1 Beyond Zero Resistance

The minimum ideas to recall are:

- below T_c the dc resistance drops to zero
- zero resistance alone does not define superconductivity
- the defining thermodynamic signature is the Meissner effect
- London theory shows that magnetic field penetrates only over a characteristic scale λ_L
- in type II superconductors, magnetic flux enters as quantized vortices between H_{c1} and H_{c2}

The two equations worth showing are:

$$\nabla^2 \mathbf{B} = \frac{\mathbf{B}}{\lambda_L^2} \qquad \Phi_0 = \frac{h}{2e}$$

Slide plan. Title: "What makes a superconductor technologically special?" Layout: left = three bullets, right = Meissner/vortex sketch. **Visual.** Use a simple comparison between a normal conductor and a superconductor, then add a small vortex sketch for type II behavior. **Advanced note.** If you want one deeper sentence, say that type II behavior is not just a theoretical curiosity: it is what makes many practical high-field superconducting materials useful in devices.

Suggested speech.

The first point I want to fix is that a superconductor is not just a perfect conductor. Zero resistance is part of the story, but it is not the full definition. The decisive signature is the Meissner effect, meaning that magnetic field is expelled from the bulk. This tells us that superconductivity is a genuine phase of matter, not simply a material with very low resistance. In type II materials, magnetic flux can enter as quantized vortices. This is not just a theoretical detail: it is exactly the regime that matters for many practical high-field devices.

Transition. A clean transition is: "So far I have described the superconducting state. The next step is to see how its collective phase becomes a device variable."

4.2 Macroscopic Phase and the Josephson Effect

The superconducting condensate is described by a macroscopic wavefunction

$$\psi(\mathbf{r}) = |\psi(\mathbf{r})|e^{i\theta(\mathbf{r})}$$

and the phase θ is physically meaningful because it controls current flow, electromagnetic response, and interference.

For two weakly coupled superconductors, the Josephson relations are

$$I_s = I_c \sin\varphi \qquad\qquad \frac{d\varphi}{dt} = \frac{2e}{\hbar}V$$

where φ is the phase difference across the junction.

Slide plan. Title: "Why phase matters" or "From phase coherence to Josephson current." Layout: a small equation block and a large Josephson junction sketch. **Visual.** A SIS-junction diagram is usually the clearest option. **Advanced note.** If you want to sound more advanced, stress that the Josephson effect turns phase into an observable electrical degree of freedom. That is the conceptual gateway from condensed matter theory to superconducting device physics.

Suggested speech.

The next step is to move from zero resistance to phase coherence. A superconductor is described not only by lossless transport, but also by a collective quantum phase. That phase matters because it can control the current. Once two superconductors are weakly coupled, the Josephson effect appears: a supercurrent can flow even at zero applied voltage, and the time evolution of the phase is tied to the voltage across the junction. This is the bridge I need for the rest of the talk, because magnetic flux can modify the phase, and the phase can then be read electrically.

4.3 SQUID as an Interferometric Flux Sensor

The central device in the talk is the **SQUID** (Superconducting Quantum Interference Device), typically a superconducting loop interrupted by one or two Josephson junctions.

For a symmetric DC-SQUID, a standard first approximation is

$$I_c(\Phi) \simeq 2I_0 \left|\cos\left(\pi\frac{\Phi}{\Phi_0}\right)\right|$$

which makes the periodic dependence on applied flux explicit.

The sentence you want the audience to remember is:

> A SQUID is an exceptionally sensitive flux-to-voltage or flux-to-current transducer.

Slide plan. Title: "SQUID: the key quantum device." Layout: large DC-SQUID loop plus a small periodic response curve. **Visual.** Make the loop dominate the slide. Keep the bullets short. **Advanced note.** In real instruments the periodic response is often linearized by operating the SQUID in a flux-locked loop, which improves dynamic range and makes the sensor easier to use as a practical magnetometer.

Suggested speech.

The SQUID is the key device because it turns quantum interference into a practical sensor. Physically, it is a superconducting interferometer. Intuitively, tiny changes in magnetic flux modify the phase around the loop, and those phase changes become measurable electrical changes. That is why SQUIDs are so useful whenever the magnetic signal is too small for conventional sensors.

4.4 The Explicit Bridge to Applications

At this point, say the bridge out loud:

- one branch uses superconductivity to sustain large currents and therefore create strong magnetic fields with low dissipation
- the other branch uses phase coherence and Josephson interference to detect extremely weak magnetic signals

Suggested speech.

Before moving to the applications, I want to make the split explicit. From the same underlying physics I get two technological directions. One direction is high current and high magnetic field with low losses, which is the route of superconducting magnets. The other is phase-sensitive quantum readout of tiny flux variations, which is the route of SQUID-based sensing. That sentence is what keeps the rest of the talk coherent.

5 Application 1: Water Treatment and Superconducting Magnetic Separation

5.1 Physical idea

This application is about the **superconducting magnet**, not the SQUID. The logic is that many pollutants are not strongly magnetic on their own, but can be attached to magnetic seeds or particles. Once magnetically tagged, they can be separated by *high-gradient magnetic separation* (HGMS).

The practical role of superconductivity is straightforward: it enables stronger fields and higher gradients with much lower losses than a purely resistive electromagnet.

5.2 What to show

- contaminant capture by magnetic particles
- flow through a high-gradient magnetic region
- retained polluted fraction versus clarified outflow
- one clear numeric callout

Slide plan. Build the slide as a left-to-right three-step process, not as a text list. **Visual.** Use an HGMS flowchart or seeded-separation sketch. A generic water photo adds little. **Advanced**

note. A useful deeper remark is that the superconducting part is not the chemistry. Its role is to make strong-field separation operationally attractive at scale.

5.3 Useful real-world number

Zhang et al. (*Cryogenics*, 2011) report COD removal up to **76%** in paper mill wastewater using modified magnetic seeds and superconducting magnetic separation. Related work has also extended the approach to challenging industrial streams such as arsenic-rich metallurgical wastewater.

Suggested speech.

The first application is deliberately chosen to show that superconductivity does not always mean quantum sensing. Here the superconducting element is the magnet, not the sensor. The workflow is simple: first, pollutants are made magnetically addressable by attaching them to magnetic particles; then the fluid passes through a region with a strong magnetic-field gradient; finally, the tagged fraction is separated from the rest of the stream. So in this case the key resource is not sensitivity, but magnetic force. The role of the superconducting magnet is to make that separation efficient and energetically realistic at high field. A concrete number to remember is the 76 percent COD removal reported by Zhang and co-workers for paper mill wastewater.

Transition. A good pivot is: "So the first example uses superconductivity as a source of strong magnetic fields. The second example moves to the opposite regime: not producing a strong field, but detecting an incredibly weak one."

6 Application 2: MEG with SQUID Arrays

6.1 Why this is a powerful example

Magnetoencephalography (MEG) is one of the clearest mature examples of SQUID technology in real clinical practice. The magnetic fields generated by neuronal currents outside the head are extremely small, roughly in the range of 10^{-15}–10^{-11} T, and measuring them requires extraordinary sensitivity.

Whole-head clinical MEG systems commonly use arrays of about **300 sensors** housed in a cryogenic dewar. The most established clinical uses include:

- pre-surgical localization of epileptic foci
- pre-operative functional mapping of eloquent cortex

6.2 What to emphasize physically

In MEG, the SQUID is not used to generate a strong field. It is used to read a very weak field directly, non-invasively, and with excellent temporal resolution. So here the key resource is not magnetic force, but extreme sensitivity.

6.3 What to show

- a real photo of a whole-head MEG system
- a simple head-plus-sensor schematic
- the field scale and the approximate number of sensors
- two concrete clinical uses

Slide plan. Use a real system image. This is the point in the talk where real hardware strongly helps credibility. **Visual.** If you include a brain activation map, it should be secondary to the actual MEG setup. **Advanced note.** If you need one more sophisticated sentence, note that MEG combines high temporal resolution with a non-trivial inverse problem, which is why anatomical constraints and signal modeling matter in clinical interpretation.

Suggested speech.

The second application is probably the strongest example of superconducting quantum sensing already embedded in real clinical workflows: MEG. Here the reason for using SQUIDs is simple and decisive. The brain produces extremely small magnetic fields, and a conventional sensor is not sensitive enough to read them reliably. SQUID arrays make that readout possible while preserving millisecond-scale temporal resolution. So in this case the key resource is extreme sensitivity to a very weak field. In practice, whole-head systems use on the order of three hundred sensors, and the best-established clinical uses are the localization of epileptic foci and functional mapping before neurosurgery. This is a very direct example of Josephson physics leaving the low-temperature lab and entering the hospital.

Transition. A useful line is: "MEG shows what SQUIDs can do in a highly controlled clinical environment. The next question is whether similar sensitivity can also be useful outside the lab and outside the hospital."

7 Application 3: Landmine and Buried-Metal Detection

7.1 Operating principle

The basic idea is **inductive metal detection**. An excitation coil produces a time-varying magnetic field, which induces eddy currents in the buried metallic target. Those eddy currents generate a weak secondary field that must be detected by the sensor.

SQUIDs are interesting here because they retain excellent sensitivity in the **low-frequency** regime. That can improve depth of investigation and help distinguish buried metallic objects relative to conventional detectors.

7.2 Why high-Tc devices matter

For field applications, high-T_c SQUIDs are attractive because they can operate around 77 K using liquid nitrogen. That usually means more noise than low-T_c devices, but much simpler logistics. This makes them especially interesting for proof-of-principle systems and specialized field deployment.

7.3 Useful real-world number

In Guo et al. (*Physica C*, 2002), a high-T_c RF SQUID metal detector readily detects an aluminum particle of only **10 mg**. The paper explicitly mentions applications to landmine searching and mobile underground-structure detection.

7.4 What to show

- excitation coil
- buried metallic target
- weak secondary magnetic field

- SQUID readout and low-frequency advantage

Slide plan. Make this a measurement schematic, not a photo-heavy slide. **Visual.** A coil-target-sensor diagram is usually the clearest choice. **Advanced note.** A good advanced comment is that sensitivity alone does not solve the full problem: background noise, false positives, mechanical robustness, and scanning strategy still matter for real deployment.

Suggested speech.

The final application changes the context again. Now I am not in a plant and not in a hospital, but in a field-detection scenario where sensitivity and operational practicality both matter. The physical principle is induced-current detection: I excite the buried metallic target, the target produces a weak secondary field, and the sensor has to read that response. The advantage of the SQUID shows up particularly at low frequency, where deeper and more selective interrogation can become possible. So here the challenge is not just sensitivity, but sensitivity in a noisy real environment. For field use, high-T_c devices are attractive because they trade some noise performance for much simpler cryogenics. A useful reference point is the detection of a ten-milligram aluminum particle reported by Guo and colleagues.

8 Comparison Slide and Closing Logic

A strong closing move is to show that the three applications are not a list of curiosities but three optimization targets built on the same platform physics.

Application	What is optimized	Role of superconductivity
Water treatment	High field and high gradient	Enables efficient magnetic separation of tagged contaminants
Clinical MEG	Extreme magnetic sensitivity	Enables direct readout of very weak biomagnetic signals with SQUID arrays
Landmine / buried-metal detection	Low-frequency sensitivity and reach	Enables detection of weak induced secondary fields from hidden targets

Slide plan. This is the cleanest comparison slide in the whole talk: no extra images, just the logic.

Suggested speech.

If I compare the three cases directly, the common structure becomes obvious. In water treatment I want strong field and strong gradient. In MEG I want extreme sensitivity to tiny magnetic signals. In buried-metal detection I want good low-frequency sensitivity in a less controlled environment. The underlying superconducting physics does not change, but the engineering objective does. That is the main message I want the audience to remember.

9 Limitations and Outlook

9.1 Shared limitations

- cryogenics still adds cost and engineering complexity
- ultra-sensitive readout demands shielding and noise control
- real devices succeed only when the full system design is robust, not just the sensor physics

9.2 Forward-looking directions

- better-performing high-T_c devices
- simpler and cheaper cryogenic platforms
- increasingly integrated low-noise electronics
- more compact sensors positioned closer to the sample or target

Slide plan. Split the slide into limits on one side and outlook on the other, then end with one large sentence across the bottom. **Advanced note.** This section is useful because it prevents the talk from sounding like pure advocacy. It shows you understand that practical impact depends on system engineering as much as on elegant physics.

Suggested speech.

To avoid ending on an overly idealized note, it is worth stating the common bottlenecks. Cryogenics remains a real constraint. Extreme sensitivity only helps if environmental noise is controlled. And in deployed systems, mechanical and electronic robustness matters as much as the physics of the sensing element. That is why the most promising directions are not only about better materials, but also about simpler cryogenics, improved low-noise readout, and more compact devices that sit closer to the actual problem.

10 Final Conclusion Ready to Deliver

Suggested speech.

To conclude, quantum superconducting devices sit at the boundary between fundamental condensed-matter physics and real technology. From the theory side, we take Meissner screening, macroscopic phase coherence, the Josephson effect, and flux quantization. From the application side, we obtain two powerful technological routes: superconducting magnets for strong-field tasks and SQUIDs for ultra-sensitive magnetic readout. The three examples in this talk, water treatment, MEG, and buried-metal detection, are therefore not isolated case studies. They are three different ways of turning the same superconducting physics into applications in environmental engineering, biomedicine, and detection technologies. In one case superconductivity creates the field. In the other, it detects the field. That is why the same physics can lead to such different technologies.

11 Speaker Checklist

Before presenting, make sure you can do these five things smoothly:

- open with a claim, not a definition
- state the split between strong-field devices and ultra-sensitive sensors early
- say explicitly what quantity is being optimized in each application
- remember the three anchor numbers: 76%, about 300 sensors, 10 mg
- close by reconnecting the applications to Meissner, phase coherence, Josephson physics, and flux quantization

12 Essential Sources

12.1 Course material

- R. Arpaia, *Lecture 4 - Superconductivity. Key Phenomena and Concepts.*
- R. Arpaia, *Lecture 7 - Superconductivity. London Equations (I).*
- R. Arpaia, *Lecture 15 - The Josephson Effect. Gateway to Superconducting Device Physics (II).*
- R. Arpaia, *Lecture 18 - Josephson Junction Devices. From Quantum Qubits to SQUIDs in Neuroscience.*

12.2 Applications

- Y. Zhang et al., *Study on industrial wastewater treatment using superconducting magnetic separation*, Cryogenics 51 (2011) 225–228. DOI: 10.1016/j.cryogenics.2010.07.002.
- J. Li et al., *FeOOH and nZVI combined with superconducting high gradient magnetic separation for the remediation of high-arsenic metallurgical wastewater*, Separation and Purification Technology 285 (2022) 120372. DOI: 10.1016/j.seppur.2021.120372.
- R. Hari et al., *IFCN-endorsed practical guidelines for clinical magnetoencephalography (MEG)*, Clinical Neurophysiology 129 (2018) 1720–1747. DOI: 10.1016/j.clinph.2018.03.042.
- X. Guo et al., *Metal detector based on high-T_c RF SQUID*, Physica C 378–381 (2002) 1404–1407. DOI: 10.1016/S0921-4534(02)01732-X.